Hostel Management System

Branch Coverage Report

Branch (decision) coverage achieved by the automated test suite

Project	Hostel Management System (HMS)
Team	Team 3
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Repository	github.com/adarshv15/hostel-management-system-team3 (commit 857e9c1)
Baseline document	Software Requirements Specification v1.0 (TEAM_3_HMS_SRS)
Report version / date	1.0 / 09-11-2026
Status	Draft for team review
Test period	Testing phase (plan: 2 Nov to 6 Nov); this run executed 09-11-2026

1. Introduction

Branch coverage checks that, for every decision in the code (an `if`, a loop test), the tests drive execution down both outcomes. A line can be fully executed while one of its outcomes never happens, so branch coverage finds gaps that statement coverage cannot. This report covers the Hostel Management System at commit `857e9c1`, measured over the 32-test automated suite.

2. Measurement Method

Item	Detail
Operating system	Linux 6.18 (cloud sandbox)
Python	3.13.18 (the project CI targets Python 3.12)
Django	5.2.19 (requirements.txt allows >=5.1,<6.0)
Coverage tool	coverage.py 7.17.2 with `--branch`
Database	SQLite, Django-managed in-memory test database
Command	`coverage run --branch --source=hostel,hms_project manage.py test -v 2`
What a branch is	coverage.py records each possible jump from a decision line to the next line (or to the exit of the function or loop). One `if` normally gives two branches.
Scope	Application code in `hostel/` and `hms_project/`. `hostel/tests.py` is excluded. Only three files contain branches: `models.py`, `admin.py` and `signals.py`.
Execution date	09-11-2026

3. Summary of Results

Measure	Result
Branches in application code	26
Branches taken by the tests	17

Measure	Result
Branches never taken	9
Decisions where only one outcome occurred (partial branches)	5
Branch coverage	65.4% (17 of 26)
Statement coverage (for comparison)	86.6% (278 of 321)
Combined figure printed by coverage.py	85.0% ((278 + 17) of (321 + 26))

Result: branch coverage is **65.4%**, well below the statement figure of 86.6%. Of the 9 missed branches, 4 are in the unused `signals.py` module, 4 are in `models.py` validation rules and 1 is in the admin bulk-delete action. The 5 partial branches (4 in `models.py`, 1 in `admin.py`) are the actionable ones: each is a decision whose second outcome no test reaches.

3.1 Branch coverage by file

File	Branches	Taken	Missed	Partial	Coverage
`hostel/models.py`	20	16	4	4	80.0%
`hostel/admin.py`	2	1	1	1	50.0%
`hostel/signals.py`	4	0	4	0	0.0%
Total	**26**	**17**	**9**	**5**	**65.4%**

All other files (`views.py`, `urls.py`, `settings.py`, `apps.py`, the migration, `asgi.py`, `wsgi.py`) contain no branches.

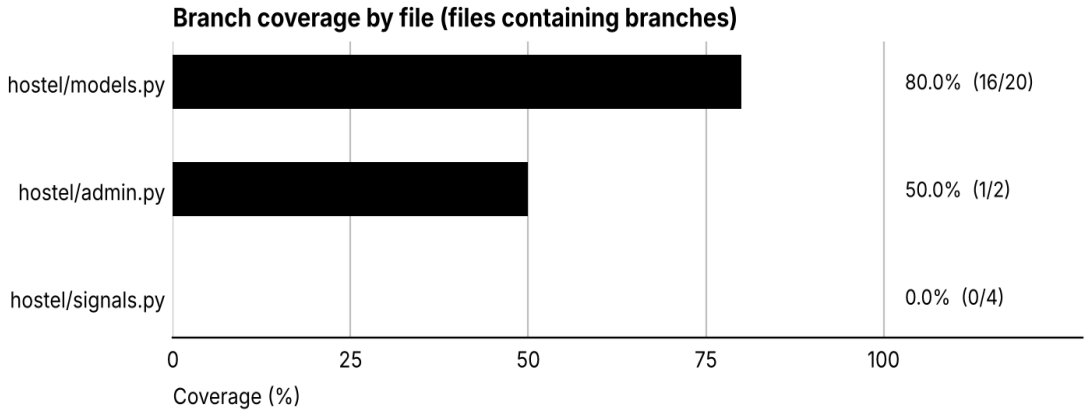


Figure 1. Branch coverage for each file that contains branches.

4. Branch-by-Branch Results

Every branch in the application code. The line numbers refer to the files at commit `857e9c1`; "exit" means the function or loop ends.

ID	Location	Decision	Outcome	Status
B-01	`models.py` 61	Active allocation for a student who is not ACTIVE?	Yes: raise ValidationError (61 to 62)	**Missed**
B-02	`models.py` 61	(same decision)	No: continue to capacity check (61 to 63)	**Covered**
B-03	`models.py` 63	Active allocation with a room?	Yes: check room capacity (63 to 64)	**Covered**
B-04	`models.py` 63	(same decision)	No: checked-out allocation or no room, skip check (63 to exit)	**Missed**
B-05	`models.py` 65	Allocation already saved (has pk)?	Yes: exclude itself from the count (65 to 66)	**Covered**

ID	Location	Decision	Outcome	Status
B-06	`models.py` 65	(same decision)	No: new allocation (65 to 67)	**Covered**
B-07	`models.py` 67	Active allocations at or above capacity?	Yes: raise "no available beds" (67 to 68)	**Covered**
B-08	`models.py` 67	(same decision)	No: allocation is valid (67 to exit)	**Covered**
B-09	`models.py` 98	Payment amount zero or less?	Yes: raise ValidationError (98 to 99)	**Covered**
B-10	`models.py` 98	(same decision)	No: continue (98 to 100)	**Covered**
B-11	`models.py` 100	Invoice and amount both present?	Yes: check against balance (100 to 101)	**Covered**
B-12	`models.py` 100	(same decision)	No: skip balance check (100 to exit)	**Missed**
B-13	`models.py` 102	Payment already saved (has pk)?	Yes: exclude itself from the paid total (102 to 103)	**Covered**
B-14	`models.py` 102	(same decision)	No: new payment (102 to 104)	**Covered**
B-15	`models.py` 109	Amount greater than balance due?	Yes: raise ValidationError (109 to 110)	**Covered**
B-16	`models.py` 109	(same decision)	No: payment is valid (109 to exit)	**Covered**
B-17	`models.py` 145	Leave end date earlier than start date?	Yes: raise ValidationError (145 to 146)	**Covered**
B-18	`models.py` 145	(same decision)	No: dates valid or missing (145 to exit)	**Missed**
B-19	`models.py` 162	Visitor exit time earlier than entry time?	Yes: raise ValidationError (162 to 163)	**Covered**
B-20	`models.py` 162	(same decision)	No: valid, or visit ongoing (162 to exit)	**Covered**
B-21	`admin.py` 81	Any audit events built for bulk delete?	Yes: bulk_create them (81 to 82)	**Covered**
B-22	`admin.py` 81	(same decision)	No: empty queryset, skip (81 to 84)	**Missed**
B-23	`signals.py` 41	Instance is an AuditEvent?	Yes: return, avoid recursion (41 to 42)	**Missed**
B-24	`signals.py` 41	(same decision)	No: create audit event (41 to 44)	**Missed**
B-25	`signals.py` 72	Another tracked model to register?	Yes: connect signals (72 to 73)	**Missed**
B-26	`signals.py` 72	(same decision)	No: loop ends (72 to exit)	**Missed**

17 of 26 branches covered.

5. Analysis of Missed Branches

ID	Location	What is not exercised	Proposed test
B-01	`models.py` 61	The rule that only ACTIVE students can hold a bed. A manual probe showed it works, but no test proves it.	Validate an allocation for a Pending, Rejected and Inactive student; expect ValidationError.
B-04	`models.py` 63	Validation of a checked-out allocation (history record) in a room that is full.	Validate a checked-out allocation in a full room; expect no error.
B-12	`models.py` 100	Payment validation when no invoice is set or the amount is missing.	Run `clean()` on a payment without an invoice; only the amount rule applies.
B-18	`models.py` 145	Leave request with valid dates passing `clean()`. The existing test only creates the record.	Run `full_clean()` on a leave with end after start, and on a same-day leave.
B-22	`admin.py` 81	Bulk delete on an empty queryset. Django normally prevents an empty selection, so this is a defensive path.	Call `delete_queryset` with an empty queryset; expect no audit events and no error.

ID	Location	What is not exercised	Proposed test
B-23 to B-26	`signals.py` 41, 72	All four branches. The module is never imported, because `HostelConfig` has no `ready()` method.	Decide: wire it in and test, or remove the module.

Adding the five tests for B-01, B-04, B-12, B-18 and B-22 would bring `models.py` to 20 of 20 and `admin.py` to 2 of 2, raising overall branch coverage from 65.4% to **84.6%** (22 of 26). Only the four `signals.py` branches would remain.

6. Limitations

- coverage.py measures jumps between lines, not each part of a compound condition. For example, line 61 combines three tests with `and`; branch coverage cannot show whether each part was true and false independently. It is weaker than condition or MC/DC coverage.
- Branch coverage shows what ran, not whether the results were asserted correctly.
- Implicit branches inside Django (ORM, admin, forms) are not part of the measurement.
- The figures come from one run on Python 3.13.18. The project's CI uses Python 3.12.
- The Test Plan's branch-coverage target was not available in the repository. If it sets a threshold, compare it with 65.4%.

7. How to Reproduce

```
coverage erase
coverage run --branch --source=hostel,hms_project manage.py test
coverage report -m --omit="*/tests.py"   # 'N->M' and 'N->exit' in Missing are branches
coverage html --omit="*/tests.py"        # partial branches are highlighted
```

8. Conclusion

The suite takes 17 of 26 branches (65.4%). The validation rules the team wrote for allocation, payment, leave and visitor data are well exercised on their failure outcome; the gaps are mostly the "valid" outcome of those rules and the unused `signals.py`. Five short tests would lift branch coverage to 84.6%.

Sign-off

Role	Name	Signature	Date
Prepared by			
Reviewed by			
Approved by (Project Guide)			

Appendix A. Raw coverage.py Output

Output of `coverage report -m --skip-empty` including `hostel/tests.py`. In the Missing column, `81->84` means the jump from line 81 to line 84 was never taken, and `100->exit` means the function exit from line 100 was never taken.

```
Name                                Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------------------------
hms_project/asgi.py                     4      4      0      0     0%   1-4
hms_project/settings.py                25      0      0      0   100%
hms_project/urls.py                     4      0      0      0   100%
hms_project/wsgi.py                     4      4      0      0     0%   1-4
hostel/admin.py                        91      8      2      1    90%   45-58, 81->84, 152-159,
169, 173-174
hostel/apps.py                          4      0      0      0   100%
```

```
hostel/migrations/0001_initial.py        9      0      0      0   100%
hostel/models.py                       155     10     20      4    92%   62, 63->exit, 71, 84,
100->exit, 115, 130, 145->exit, 149, 166, 180, 191, 206
hostel/signals.py                       17     17      4      0     0%   2-80
hostel/tests.py                        205      0      2      0   100%
hostel/views.py                          8      0      0      0   100%
------------------------------------------------------------------------------
TOTAL                                  526     43     28      5    91%

3 empty files skipped.
```